

Comprehensive University Management System

Manage the data ecosystem of an academic institution through a robust architecture.

The objective is to centralize information across the university community, control academic processes such as enrollments and campus transfers, and automate the output of official information for printing, while maintaining strict referential integrity.

FUNCTIONAL REQUIREMENTS

- **User Management (Hierarchy):** Register differentiated profiles by applying inheritance. A *Persona* superclass is defined to encapsulate shared attributes (name, CURP/Tax ID, address), from which specialized subclasses derive: *Student*, *Employee*, *Confidential Staff*, and *Administrative Staff*, each containing its own specific attributes and methods in code.
- **Academic Module:** Process the business logic of student academic trajectories. This includes enrolling students, validating course prerequisites via SQL queries, managing capacity during schedule assignments, and formally registering campus or faculty transfers.
- **Dynamic Reports:** Export processed information into physical formats. The system will extract data and dynamically generate Rich Text Format (RTF) files formatted to official standards for enrollment certificates and print-ready academic transcripts.

TECHNICAL IMPLEMENTATION NOTES

- **Database:** The model must originate from a solid EER diagram designed in MySQL Workbench. Table creation (DDL scripts) must guarantee the proper use of primary and foreign keys to relate the *Persona* table with its derived entities.
- **Interface & Persistence:** The GUI (e.g., developed using Java Swing with multiple windows) will act as the controller. In case of SQL server disconnects, the system can rely on a hybrid persistence mechanism using local .txt files temporarily.

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